

MAI - A Proactive Speech Agent for Metacognitive Mediation in Collaborative Learning

JUSTIN EDWARDS, Learning and Educational Technology Lab, University of Oulu, Finland

ANDY NGUYEN, Learning and Educational Technology Lab, University of Oulu, Finland

MARTA SOBOCINSKI, Learning and Educational Technology Lab, University of Oulu, Finland

JONI LÄMSÄ, Learning and Educational Technology Lab, University of Oulu, Finland

ADELSON DE ARAUJO, University of Twente, the Netherlands

BELLE DANG, Learning and Educational Technology Lab, University of Oulu, Finland

RIDWAN WHITEHEAD, Learning and Educational Technology Lab, University of Oulu, Finland

ANNI-SOFIA ROBERTS, Learning and Educational Technology Lab, University of Oulu, Finland

MATTI KAARLELA, Learning and Educational Technology Lab, University of Oulu, Finland

SANNA JÄRVELÄ, Learning and Educational Technology Lab, University of Oulu, Finland

We introduce MAI - a proactive speech agent aimed at enhancing metacognitive awareness among learners in collaborative learning settings. Background is presented around Socially Shared Regulation of Learning and the role of metacognition in learning. Next, the design of the rules that MAI uses to prompt learners and mediate metacognition are introduced. We describe the conditions in which MAI has been piloted thus far, including as a Wizard of Oz prototype and as a fully functional prototype using natural language processing. We discuss the ethical considerations that went into the prototyping and testing of MAI. Finally, we describe our next steps for understanding the interactions learners had with MAI already, planned design changes, and the future of testing the agent.

CCS Concepts: • **Human-centered computing** → Empirical studies in HCI; Empirical studies in interaction design; **Natural language interfaces**; • **Applied computing** → **Collaborative learning**.

Additional Key Words and Phrases: proactive agents, speech agent, speech interfaces, collaborative learning, regulation of learning

ACM Reference Format:

Justin Edwards, Andy Nguyen, Marta Sobocinski, Joni Lämsä, Adelson de Araujo, Belle Dang, Ridwan Whitehead, Anni-Sofia Roberts, Matti Kaarlela, and Sanna Järvelä. 2024. MAI - A Proactive Speech Agent for Metacognitive Mediation in Collaborative Learning. In *ACM Conversational User Interfaces 2024 (CUI '24)*, July 8–10, 2024, Luxembourg, Luxembourg. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3640794.3665585>

1 INTRODUCTION

In order to accomplish their learning goals, students must coordinate various cognitive activities, including making plans, selecting strategies, monitoring their progress, and adapt their behaviour, in a process of self-regulation of learning [21]. Success in collaborative learning contexts requires regulation as well - learners must not only self-regulate, but also co-regulate the learning of peers, and engaging in group-level, socially shared regulation of learning (SSRL) by which a group deliberately plans, enacts strategies, reflects, and adapt their shared strategies and beliefs [10]. SSRL

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

© 2024 Copyright held by the owner/author(s).

Manuscript submitted to ACM

is driven by metacognitive processes of monitoring the cognitive, behavioural, and motivational states of the group, evaluating these states against joint goals and standards, and collectively engaging in negotiated adaptations [10]. Indeed research on collaborative learning has shown that SSRL processes contribute to group success in these learning contexts [9]. The metacognitive processes of monitoring, evaluating, and implementing adaptations are paramount for successful collaborative learning, but little progress has been made in facilitating these complex group level activities during collaborative learning.

One promising area for supporting SSRL during collaborative learning is the concept of trigger events. Recent collaborative learning research has introduced the concept of trigger events - moments in which these group-level SSRL processes are required as strategies for responding to challenges like disruptions to learning, or negative evaluations of learners' ongoing conditions [8]. Evidence for these trigger events has largely come from studies of physiological data during collaborative learning activities, stemming from the observation that group members in a collaborative learning setting synchronise in their physiologically responses during the metacognitive activities of SSRL [7]. Recent work has proposed that other streams of data indicating groupwide synchrony may also offer evidence of these group-level metacognitive activities which follow trigger events [17, 20]. These multimodal indications of trigger events offer a window into supporting metacognitive processes, as detection of these triggers in real-time would enable an observer (be it a teacher or a digital support tool) to detect that learners may be facing a groupwide challenge requiring regulation and respond by supporting SSRL via prompts to raise group-level metacognitive awareness and engage in the monitoring, evaluating, and implementation of adaptations needed to overcome that challenge [8].

2 INTRODUCING MAI

We introduce MAI, a proactive speech agent for collaborative learning which acts as a metacognitive mediator [15]. In this role, MAI listens to the conversation between members of a group in a collaborative learning task and seeks to detect trigger events using rules based in SSRL literature. When a trigger event is detected, MAI intervenes in the interaction with speech, prompting students to raise group-level metacognitive awareness through reflective questions. MAI's rules are as follows:

- **NEGATIVE VALENCE MONITORING COGNITION:**
 - IF multiple group members use keyword "[I/we] don't../can't/aren't" on 3 consecutive turns THEN prompt one of the following:
 - "I noticed that you are facing a difficulty ...
 - * ... Can you describe what is going on right now? Do you all agree?"
 - * ... What might clarify this concept/issue for you?"
 - * ... How could you help yourself progress?"
 - * ... How is your current approach helping you achieve your goals?"
- **ISSUE RELATED TO CONCEPTUAL UNDERSTANDING**
 - IF multiple group members express confusion related to the same target (i.e., domain knowledge) in one episode, over 3 consecutive turns, THEN prompt one of the following:
 - I noticed that you might be confused ...
 - * ... Can you explain what are you trying to understand now?"
 - * ... Do you all understand your task the same way? How about the key concepts?"
 - * ... How confident do you feel that you have the tools to complete your task? What would help you?"

- LACK OF SHARED PERSPECTIVE:
 - IF multiple group members use of first or second person singular ("I"/"You") over three consecutive turns. THEN prompt one of the following:
 - "I noticed that you are thinking of your individual perspectives ...
 - * ... How can you progress as a group?"
 - * ... What is your joint plan for solving the task?"
 - * ...What are you trying to accomplish here as a group?"

These rules are followed in the priority order that they are listed - that is to say, if the negative valence monitoring cognition rule and the lack of shared perspective rule are both triggered in the same turn, an intervention from the negative valence monitoring cognition rule list is selected. Additionally, interventions are not to occur within the first five minutes of the interaction (to allow groups to familiarise themselves with the task and each other without prompting), and after an intervention, 3 minutes must elapse before another intervention can be triggered (so that interventions were not overwhelming or focused on the same trigger event). These rules were co-designed between the researchers and a lecturer from the teacher training course from which pilot study participants were sampled.

Participants were instructed that MAI could hear them and would intervene with speech via a tablet on their table which had its screen turned off. In our first pilot study of MAI, we used a Wizard of Oz design, by which researchers listened to groups' conversations and then played a pre-recorded, randomly selected intervention from the list. Interventions were recorded by a Finnish speaker. When a rule was triggered, Wizards in another room played an audio file, using the tablet as a remote speaker.

In our second pilot study of MAI, participant speech was transcribed using Whisper (large-v3 model [16]) with speaker diarization labelled using [1] on a Tesla P100 GPU accessed on a virtual machine through the university's high performance cluster. Transcribed and diarized speech was sent to the configurable text chatbot Clair [3] using the rules specified above for dialogue management. When an intervention was triggered, a response from the chatbot would automatically play an audio file of the recorded intervention, using the tablet as a remote speaker. Interventions in the second pilot were recorded by the same person speaking Finnish. Code for the transcription and diarization of participant speech and for sending speech to Clair and receiving audio responses is provided¹.

Each pilot aimed to answer different questions about the design and implementation of MAI. The aims of the first pilot were to understand whether metacognitive mediation interventions following trigger events detected through learners' speech interactions are effective in facilitating SSRL (RQ1). The aims of the second pilot were to understand whether MAI could be implemented in a prototype form without the need of a human wizard for natural language processing (RQ2).

3 METHODS

3.1 Participants

Pilot 1: A sample of 52 Finnish speaking pre-service teachers was recruited from a Finnish university's teacher education program. With the consent of a lecturer from the course, all students from a module on Intercultural Teacher Education were invited to participate in the study. Participants were briefed on the aims of the study and the data collection methods. The study took place during one scheduled 90 minute class meeting of the students' module, and all students in the course engaged in the learning activity, with those who opted out of data collection engaging in the same

¹Author github page removed for anonymous review

collaborative learning activity but without any data collection or interactions with MAI. The study received ethical approval through the university's ethics procedures for human participant studies (Ethics code UMLA-AI-PRCL).

Pilot 2: A sample of 11 additional English-speaking pre-service teachers was recruited from the same program. With the consent of a lecturer from the course, all students from the English-taught offering of the Intercultural Teacher Education module were invited to participate in the study. The study took place during one scheduled 90 minute class meeting of the students' module, and all students in the course engaged in the learning activity, with those who opted out of data collection engaging in the same collaborative learning activity but without any data collection or interactions with MAI. The study received ethical approval through the university's ethics procedures for human participant studies (Ethics code UMLA-AI-PRCL-Spring).

3.2 Materials

3.2.1 Learning Task. We created a collaborative learning script which consisted of an introductory video to the topic (5 min) [11], a simulation with a set of guiding questions and answer sheet (20 min) [14]; and a hands-on experimental task that focused on the energy forms and energy conservation when different balls bounced on the floor, which we co-designed with the teacher of the module (25 min). Groups of students progressed through the script at their own pace in a GoLab virtual learning environment² on a group's shared screen, controlled by a mouse and keyboard on their table.

3.2.2 Reflection Task. After completing the hands-on physics task, students were provided markers, sticky notes, and a poster paper with two questions written on it: "reflect on your thoughts regarding the use of simulations and 'hands-on' experiments in education" and "How can a student be aided in connecting the phenomena and concepts explored in simulations to other experimental, 'hands-on' work?." In this way, the learning task had both a physics content focus, but also a focus on teaching physics and learning design, so that it would be a beneficial part of our participants' teacher training module.

3.2.3 MAI Interventions. MAI prompts were recorded by a non-binary Finnish person who is fluent in both Finnish English, and MAI was not referred to with gendered language so as to minimise cues of the agent's gender. This decision follows from recent work in CUI literature challenging the design tendency of speech agents being gendered as female (and being explicitly cisgendered at all) by default [5, 18, 19]. Interventions were proceeded by a 1.5 second long chime sound and then 1.5 seconds of silence, building on research around proactive speech agents which suggests that proactive speech forewarned by a request for social access may be more readily accepted by a listener [6]. All interventions are described above and lasted between 8 and 12 seconds, inclusive of the chime and silence. Student groups heard the recorded intervention audio through an iPad at maximum volume in the center of their table with its screen locked and display off.

3.3 Measures

3.3.1 Partner Model Questionnaire. Participants were asked to complete the 18-item Partner Model Questionnaire (PMQ). The PMQ is a validated self-report scale consisting of word pairs separated by a 7-point semantic differential scale. The scale comprises three subscales: *partner competence and dependability*, *human-likeness*, and *communicative*

²<https://www.golabz.eu/>

flexibility onto which nine, six, and three items load respectively [4]. Participants completed the PMQ before and after interacting with MAI.

3.3.2 Domain Knowledge Tests. Before and after the experiment, participants completed a domain knowledge test evaluating their understanding of physics concepts which were central to the collaborative learning task.

3.4 Physiological Measures

During the experiment, electrodermal activity (EDA) and heart rate data were collected with physiological Shimmer GSR3+ sensors. The sensors were synchronized before each session and attached to the participants' non-dominant hands so that gel electrodes were placed on the thenar and hypothenar eminences on their palms [2] and on their earlobe on the same side of the body. The sampling rate of the signals was set to 128 Hz.

3.5 Video and Audio Recording

Video and audio recordings were made of each group using wide-angle Konftel Cam20 conference cameras and individually-worn microphones in order to capture all speech, gestural, and facial expression data throughout the experiment.

3.6 Interviews

After the experiment, groups engaged in a self-directed semi-structured interview by discussing aloud two questions: "How did your collaboration go?" and "How do you think MAI (an AI agent) has been useful or not useful in your collaboration?". Questions were presented on a sheet of poster paper with the instructions to discuss both questions aloud, and participants were free to discuss for as long as they wanted.

3.7 Procedure

Pilot 1: Participants were provided information sheets, consent forms, and data privacy letters in a class meeting before the experiment. On the day of the experiment, students were arranged into groups of 4 in separate rooms, with students who did not consent to participate in data collection still joining in the learning task, but without any of the data collection materials. An informational video was shown explaining the task and researchers helped students attach microphones and Shimmer devices. After all students in a group were ready to begin, researchers distributed the self-report measures to each group. When all group members finished filling out these measures, the group was invited to begin progressing through the learning script. Students engaged with the learning script at their own pace, with interventions from MAI when triggered by an intervention rule. MAI also intervened to inform students when there were 5 minutes remaining in each of the simulation and hands-on tasks. At the end of the hands-on task, researchers distributed poster paper and post-it notes so participants could engage in the reflection task. The reflection task lasted for as long as students wanted typically for about 10 minutes. When groups were finished the reflection task, researchers distributed the self-report questionnaires again. After all group members finished these, researchers distributed a large sheet of paper with the two interview questions written on it and instructed students to discuss the questions for as long as they wanted, with interviews typically lasting about 5 minutes. When discussions ended, researchers helped participants to remove the recording devices, debriefed them, and thanked them for their participation. All learning materials and questionnaires were either developed in Finnish or translated to Finnish (in the case of the partner model questionnaire).

Pilot 2: The procedure for the second pilot was the same as the first, with students in two groups of four and one group of three in separate rooms. All learning materials and measures were translated from Finnish to English (or presented in their original English form) to accommodate the fact that students from the second pilot came from an English speaking module and were not all fluent in Finnish. Because of computational resource constraints, only one group in Pilot 2 (the group of three students) interacted with the prototype, fully automated NLP version of MAI described above. One of the other groups interacted with the Wizard of Oz version of MAI described above, while the other did not interact with MAI at all. This allowed for comparison between the prototype MAI and Wizard of Oz MAI, to better understand RQ2, as well as a comparison between students who engaged in the learning task with MAI and those who engaged in the task without MAI to better understand RQ1.

4 ETHICAL CONSIDERATIONS

Because a number of sensitive data types (physiological, video, and audio data) were collected for these pilots, we were especially sensitive to ethical concerns in collection and analysis of data. Most notably, because Pilot 2 involved the transcription of speech using an AI model, we aimed to minimize the potential harm to participants by hosting the model locally on a university server rather than using API calls to access the model and transmitting participant speech data over the internet. This choice ensures that, just as with data analysis, all data pre-processing during collection remains entirely and solely in the control of the researchers. This further allowed the researchers to assure participants that their data would not be used to train any AI model nor would it be retained in any form by any model - an assurance we could not make as definitively had we used non-local version of Whisper. This decision was not without tradeoffs, as the computational resources required to run the model locally are large and restricted the number of participants who could interact with the prototype version of MAI simultaneously. That said, we believe that committing to data minimisation through the use of local models increases participant data privacy, reduces the reliance of publicly funded research on private sector organisations, and increases participant enthusiasm for scientific participation, in part contributing to the large sample we were able to recruit.

5 NEXT STEPS

The pilot studies with MAI are works in progress, with data fully collected but not yet analysed for either study. Planned analysis includes qualitative coding of metacognitive monitoring using the video data and scoring of domain knowledge pre and post-tests to understand the effect of MAI interventions on both SSRL processes and learning outcomes. Likewise, Partner Model Questionnaire and interview results will be analysed to understand student impressions of MAI as both a conversational partner and as a tool for supporting collaboration. Audio transcripts and physiological measures will likewise be analysed to better understand relationships between the trigger moments which were detected by MAI and trigger moments that would be indicated by physiological and linguistic data, building on new work combining these data sources for detecting triggers [12].

Initial data familiarisation revealed participant disinterest when an intervention starts with phrasing which is repetitious of a previous intervention, so we intend to implement new interventions which vary phrasing, seeking to increase impressions of communicative flexibility, drawing on literature around partner models machine dialogue partners [4].

Data analysis will be aimed at investigating a number of questions regarding the design of MAI. Insofar as MAI is aimed to invited SSRL in students, data modalities including physiological measures and discourse between group members will be analysed to compare whether patterns known to be associated with regulatory processes (see [7, 17, 20])

manifested following MAI's prompts. Qualitative coding of the learning processes which proceeded and followed MAI prompts will give insights into how effective each category of prompting was, both in targeting the intended trigger events and in inviting regulatory processes. This analysis will allow further iterative design of MAI, with the aim of accurately identifying trigger events via the design of prompting rules, and effectively inviting regulation with the design of prompts.

In terms of technical implementation, a number of issues still remain to be solved. Whisper is prone to perceptual artefacts, whereby non-speech noise or silence is interpreted as speech, sometimes leading to a self-perpetuating loop of imagined speech [13]. In our pilot of MAI using Whisper, these loops of imagined speech required a manual reset of the Whisper model by a researcher, causing MAI to "not hear" participants for about 60 seconds while the model restarted. Further modifications to room and microphone arrangement, Whisper hyperparameters, and underlying speech recognition architecture may help to mitigate these issues in the future, but errors in realtime speech recognition like this are inherent to even state-of-the-art approaches [13]. Likewise, while diart represents a state-of-the-art approach to diarization, the low-latency implementation of the model needed for real-time diarization in this context still benchmarks at a diarization error rate about 20% [1]. Due to both of these sources of potential error, close analysis of the logs of the transcripts which were fed to the Clair NLP system are needed, as both false-positive and false-negative interventions from MAI may have occurred. These issues are endemic to the pursuit of real-time speaker-aware natural language processing at the moment, so the utility of implementation of MAI using this technology rather than human wizards is dependent on both the sensitivity that researchers have to these intervention errors as well as the tradeoff of the cost of training and using human wizards instead.

Following data analysis and further refinement of prompts, we aim to implement a prototype of the agent in secondary school classrooms. Insofar as the learning task we selected is appropriate for the 7th grade Finnish science curriculum, we would want to test the same procedure (without the reflective task aimed at teaching training students) in an actual classroom with students at that level. Because of the computational limitations described above, this implementation will likely utilize both the prototype and Wizard forms of MAI. While data from the pilot studies is still being cleaned and analysed, the technological prototype of MAI seemed to broadly select the moments for prompting which matched prompting patterns of Wizards, who performed according to the basic prompting rules without error. This is a promising sign for the use of bot prototypes side-by-side, but further comparison of differences in prompting and costs is needed. Further exploration of ethical access to more powerful computational resources will also impact next steps with MAI.

ACKNOWLEDGMENTS

This project was financially supported by the Research Council of Finland (formerly known as Academy of Finland) Profi7 352788 (Hybrid Intelligence) and grants 355776 (LEAD) and 350249 (UMLA-AI-PRCL). Special thanks to Laura Timonen for helping us to design the learning task and prompting rules and along with Essi Lakso integrating this study into their courses. Thanks to Pantelis Papadopoulos, Shant Derounian and Jakob Sikken for technical and research design support. And heartfelt thanks to our Wizards and supporting researchers in LET - Antti, Tiina, Ahsen, Sara, Eija, Faith, Tam, Boshra, and Tiia as well as the pre-service teachers who participated in our study.

REFERENCES

- [1] Juan M. Coria, Hervé Bredin, Sahar Ghannay, and Sophie Rosset. 2021. Overlap-Aware Low-Latency Online Speaker Diarization Based on End-to-End Local Segmentation. In *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. 1139–1146. <https://doi.org/10.1109/ASRU51503.2021.9688044>
- [2] Michael E. Dawson, Anne M. Schell, and Diane L. Filion. 2007. *The Electrodermal System*. Cambridge University Press, 159–181.

- [3] Adelson de Araujo, Pantelis M Papadopoulos, Susan McKenney, and Ton de Jong. 2023. Supporting Collaborative Online Science Education with a Transferable and Configurable Conversational Agent. *CSCL23* (2023).
- [4] Philip Doyle. 2022. *The Dimensions and Adaptation of Partner Models in Human-Machine Dialogue*. PhD Thesis. University College Dublin. School of Information and Communication Studies.
- [5] Justin Edwards, Leigh Clark, and Allison Perrone. 2021. LGBTQ-AI? Exploring Expressions of Gender and Sexual Orientation in Chatbots. In *Proceedings of the 3rd Conference on Conversational User Interfaces* (Bilbao (online), Spain) (CUI '21). Association for Computing Machinery, New York, NY, USA, Article 2, 4 pages. <https://doi.org/10.1145/3469595.3469597>
- [6] Justin Edwards, Christian Janssen, Sandy Gould, and Benjamin R Cowan. 2021. Eliciting spoken interruptions to inform proactive speech agent design. In *Proceedings of the 3rd Conference on Conversational User Interfaces*. 1–12.
- [7] Eetu Haataja, Jonna Malmberg, and Sanna Järvelä. 2018. Monitoring in collaborative learning: Co-occurrence of observed behavior and physiological synchrony explored. *Computers in Human Behavior* 87 (2018), 337–347.
- [8] Sanna Järvelä, Andy Nguyen, and Allyson Hadwin. 2023. Human and artificial intelligence collaboration for socially shared regulation in learning. *British Journal of Educational Technology* (2023).
- [9] Hanna Järvenoja, Sanna Järvelä, and Jonna Malmberg. 2020. Supporting groups' emotion and motivation regulation during collaborative learning. *Learning and Instruction* 70 (2020), 101090.
- [10] Sanna Järvelä, Allyson Hadwin, Jonna Malmberg, and Mariel Miller. 2018. Contemporary Perspectives of Regulated Learning in Collaboration. In *International Handbook of the Learning Sciences* (1 ed.), Frank Fischer, Cindy E. Hmelo-Silver, Susan R. Goldman, and Peter Reimann (Eds.). Routledge, New York, NY : Routledge, 2018., 127–136. <https://doi.org/10.4324/9781315617572-13>
- [11] Peekaboo Kidz. 2015. Energy. <https://www.youtube.com/watch?v=Q0LBegPWzrg>
- [12] J. Lämsä, J. Edwards, E. Haataja, M. Sobocinski, A. Nguyen, P.R. Peña, and S. Jarvela. submitted. Learners' linguistic alignment and physiological synchrony: Identifying trigger events that invite socially shared regulation of learning. *Journal of Learning Analytics* (submitted).
- [13] Ashish Mittal, Rudra Murthy, Vishwajeet Kumar, and Riyaz Bhat. 2024. Towards understanding and mitigating the hallucinations in NLP and Speech. In *Proceedings of the 7th Joint International Conference on Data Science & Management of Data (11th ACM IKDD CODS and 29th COMAD)*. 489–492.
- [14] University of Colorado Boulder. 2023. Energy skate park. <https://phet.colorado.edu/en/simulations/energy-skate-park>
- [15] Jun Oshima and Ritsuko Oshima. 2013. Collaborative Learning Through Socially Shared Regulation Supported by a Robotic Agent. In *To See the World and a Grain of Sand: Learning across Levels of Space, Time, and Scale: CSCL 2013 Conference Proceedings Volume 2*. Madison, WI, 327–328. <https://doi.org/10.22318/cscl2013.2.327>
- [16] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning*. PMLR, 28492–28518.
- [17] Márta Sobocinski, Sanna Järvelä, Jonna Malmberg, Muhterem Dindar, Antti Isosalo, and Kai Noponen. 2020. How does monitoring set the stage for adaptive regulation or maladaptive behavior in collaborative learning? *Metacognition and Learning* 15 (2020), 99–127.
- [18] Selina Jeanne Sutton. 2020. Gender Ambiguous, not Genderless: Designing Gender in Voice User Interfaces (VUIs) with Sensitivity. In *Proceedings of the 2nd Conference on Conversational User Interfaces*. ACM, Bilbao Spain, 1–8. <https://doi.org/10.1145/3405755.3406123>
- [19] Suzanne Tolmeijer, Naim Zierau, Andreas Janson, Jalil Sebastian Wahdatehagh, Jan Marco Marco Leimeister, and Abraham Bernstein. 2021. Female by Default? – Exploring the Effect of Voice Assistant Gender and Pitch on Trait and Trust Attribution. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–7. <https://doi.org/10.1145/3411763.3451623>
- [20] Eija Vuorenmaa, Sanna Järvelä, Muhterem Dindar, and Hanna Järvenoja. 2023. Sequential patterns in social interaction states for regulation in collaborative learning. *Small Group Research* 54, 4 (2023), 512–550.
- [21] Barry J Zimmerman. 2002. Becoming a Self-Regulated Learner: An Overview. *THEORY INTO PRACTICE* (2002).